

(a) Collection

Collector

seed addresses,
explorer APIs,
cached raw JSON

Decoder

transfer, bridge,
swap, lending $\rightarrow$
canonical events

(b) Trajectory

Trajectory builder

bounded value-
flow expansion;
bridge-link confidence

Prefix features

temporal, structural,
economic, semantic-
action, motif

(c) Early detection

Scorer M1

tree ensemble
on $\mathbf{x}(\tau_{1:k})$;
Platt calibration

Alert + evidence

$\hat{p}_k \geq \theta$; top
features and motifs